

Jiayi (Johnny) Li

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EDUCATION

Northwestern University

B.S. Major in Computer Science and Statistics

Evanston, IL

09/2025 - 06/2029 (Expected)

- GPA: 4.0/4.0

- Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Intro to Stats and Data Science

WORK EXPERIENCE

Lendo Capital

New York City, USA

Quantitative Analyst Intern

08/2026 - Present

- Built a Python (pandas, yfinance) pipeline with parquet caching to ingest and clean 5 years of OHLCV data across 51 tickers, fixing a bug that let invalid rows silently inflate coverage, with per-step audit logging
- Implemented return and portfolio analytics in Python (arithmetic/log returns, rolling volatility, Sharpe ratio) across 50 equities, validated to floating-point precision against hand-calculated values
- Engineering look-ahead-safe value and momentum factor signals in Python with lagged cross-sectional ranking, building a reusable quintile-sort function for long-short backtests

Sunnybrook Hospital

North York, Canada

Software Engineering Intern

01/2026 - 09/2026

- Built a Selenium automation pipeline in Python using WebDriverWait, XPath, and JavaScript DOM event dispatch to programmatically run SIMAC2 beam simulations on the Linux platform
- Developed a Python tool using NumPy and PyVisa that fits linear regressions to pre/post-irradiation voltage drifts to extract absorbed dose, cutting analysis from ~30 minutes to under a minute
- Built a multithreaded PyVisa/SCPI acquisition script over GPIB applying the Callendar-Van Dusen equation across up to 6 RTD probes for real-time sub-second temperature logging

My Learning Space AI (MYLS)

Toronto, Canada

AI Product Manager Intern

06/2025 - 05/2026

- Conducted A/B testing and user interviews to identify key friction points, designed and iterated an optimized booking flow improving successful bookings by 9%, and drove data-backed improvements in monetization
- Created AI-generated mock prototypes and collaborated cross-functionally with engineering and design to support early-stage user research and implement data-backed product improvements
- Co-authored a 5-chapter book published by China Financial & Economic Publishing House on AI's structural impact on education and institutional decision-making in North America

PROJECTS

Morning Brief Bot – Python

- Built a fault-tolerant pipeline integrating 4 APIs (Open-Meteo, Google Calendar via OAuth2, 9 RSS/Google News feeds), with per-source exception isolation so one failed API never breaks the digest
- Wrote a regex-based keyword filter (word-boundary matching, 20+ terms) to extract relevant articles from general news feeds, with URL deduplication across overlapping queries
- Deployed as a serverless GitHub Actions cron job with encrypted repo secrets, rendering an HTML email via Jinja2 and delivering it over Gmail SMTP

Grademaster – Python

- Built a full-stack grade tracking web app in Python and Flask supporting weighted items and grouped sub-components, with a REST API that auto-saves edits on a 500ms debounce
- Engineered a what-if solver that isolates any pending assessment as a variable and solves for the exact score needed to hit a target final grade, holding all other pending items at their predicted values
- Deployed via Unicorn on Render with atomic JSON persistence using temp-file-replace writes, requiring only pip install -r requirements.txt with no database or external storage needed

S&P 500 Market Crash Simulator – Python

- Led a 4-person team building a financial contagion simulator across 503 S&P 500 companies using a weighted graph of ~24,000 Pearson correlation edges ($r \geq 0.7$) from COVID-19 period log-returns
- Engineered a recursive contagion algorithm in Python attenuating crash impact across correlated neighbors, simulating user-defined crash scenarios with real-time market cap loss estimates

- Built a full-stack interactive web dashboard using Dash and Plotly, integrating yfinance historical data retrieval, NetworkX graph layout computation, and NumPy correlation analysis into a single deployable pipeline

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, HTML/CSS, Javascript, R
- **Frameworks:** Flask, Vue, React, Dash, PyVisa
- **Libraries:** Plotly, NumPy, Pandas, Matplotlib, SciPy, SciKit-Learn, PyTorch
- **Developer Tools:** Git, VS Code, PyCharm, Microsoft Offices, Render, Github